



Measurement of leaf surface area

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All opinions expressed in the report reflect only the views of the authors.

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Introduction

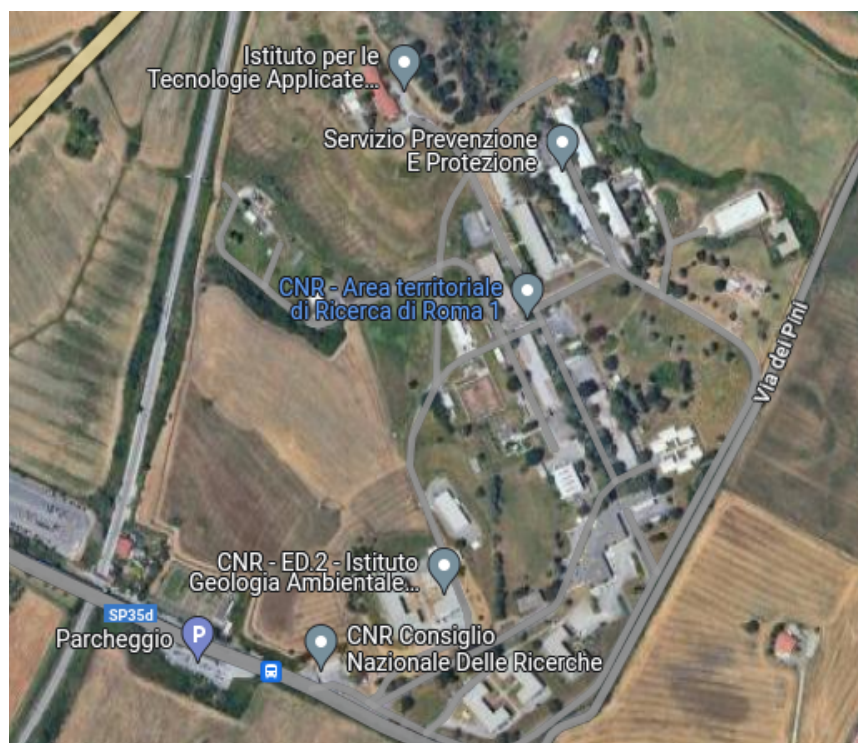
Measuring leaf surface area is a crucial tool in plant research¹. It is an important indicator of plant photosynthetic capacity, or to evaluate water loss through evapotranspiration. It can also be used to assess the release of volatile organic compounds (VOCs) or the absorption of airborne substances. These insights are valuable for agriculture, forestry, and environmental monitoring efforts.

National Agency for the Evaluation of the University and Research System (ANVUR) defines the third mission as "propensity of structures towards openness towards the socio-economic context, exercised through the valorisation and transfer of knowledge"²

In line with the Third Mission of universities and research institutions to share scientific knowledge with the public, the CNR Montelibretti Research Area in Rome (42.10576N, 12.63737E) has organized an OpenDay event for secondary school students. This hands-on learning experience will introduce students to the scientific method through the measurement of leaf surface area.

By actively participating in the key steps of the scientific process, from sampling leaves to analyzing results, students will gain valuable experience in critical thinking, problem-solving, and data interpretation. This activity has the potential to spark students' interest in science, technology, engineering and mathematics (STEM) fields and inspire them to pursue further scientific exploration.

This guide provides teachers with a detailed description of the activity, including discussion questions and a list of readily available materials, allowing students to repeat the experiment at school or home.



CNR - Rome 1 Research Area

1. SAMPLING

Treccani definition - Sampling

Statistical survey strategy which consists in the observation of a part of the units that make up a population in order to obtain valid information for the entire population. The subset of units on which the survey is carried out is called the sample.

Minimum number of samples to have a standard deviation → 2

$$\text{Deviazione Standard} \rightarrow s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{(n-1)}}$$

Minimum number of safety samples → 3

The sample should be as representative of the population as possible.

2. IMAGE ACQUISITION

It is possible to acquire the image in different ways:

1. With a camera → SD → computer
2. With a smartphone → (cloud, mail, etc.) → computer
3. With a webcam → computer
4. With a scanner → computer

It is essential to have a dimensional reference in the photo, an image whose dimensions are known (Fig.1)

3. MEASUREMENT – DATA

Treccani Definition - Measurement

The numerical value attributed to a quantity, obtained and expressed as the ratio between the given quantity and another of the same kind taken as a unit (unit of measurement), and determined with appropriate methods or measurement instruments.

The measurement of a physical quantity is an approximation of the true value of the physical quantity.

Accuracy of a measurement

Accuracy means the extent of the difference between the measured value and the true value. The smaller the difference, the greater the accuracy of the measurement.

Precision of a measurement

The precision of a measurement means the extent of the difference between the measurements compared to the average value. The smaller the magnitude of the difference the greater the precision.

To measure the surface of a leaf we can use different methodologies.

A. We can draw the leaf on a sheet of graph paper and count the number of squares enclosed in the drawing (Fig.2).

B. We can draw the leaf on a white sheet of paper and simultaneously draw a geometric figure (for example a square) of known area. Subsequently, the two figures are cut out and weighed (Fig.3). With a simple mathematical ratio it is possible to trace the surface of the "leaf of paper" and therefore of the leaf.

$$A_{leaf} = A_{geometric\ square} * \left(\frac{W_{paper\ leaf}}{W_{paper\ geometric\ square}} \right)$$

A = Surface area

W = Weight

C. We can take a digital image of the leaf and use software to measure its surface.

Following this last method to "measure" a digital image we can use the open source ImageJ software which can be used on the three most popular operating systems.

- Download from the site <https://imagej.net/ij/>
- Extract the contents of the compressed file and run it
- File → Open Image
- Define the measurement scale
 - Select → Straight Line
 - Draw a line on the reference image (Fig.4)
 - Select → Analyze → Set Scale (Fig.5)
- Set the Color Threshold
 - Select Image → Adjust → Color Threshold (Fig.6)
 - Modify Hue; Saturation; Brightness e deselectare Dark background
 - Select
- Measure the surface area
 - Select Analyze → Analyze particles (Fig.7)
 - Set Show → Outlines
 - Select → Display results; Clear results; Include holes
 - Select → Ok
- Save table Results

4. ANALYSIS AND RESULTS

If the leaf sample is representative of all the leaves of the plant it is possible to determine the overall leaf surface area of the plant.

$$A_c = \sum_{j=1}^n a_j$$

$$A_T = A_c * N_p$$

A_c = Sample Leaf Surface

N_p = Total number of leaves Plant

A_T = Total Leaf Area

REFERENCES

1. <https://www.sciencedirect.com/science/article/pii/S2351989420306703>
2. <https://www.anvur.it/en/activities/third-mission-impact/>

APPENDIX 1

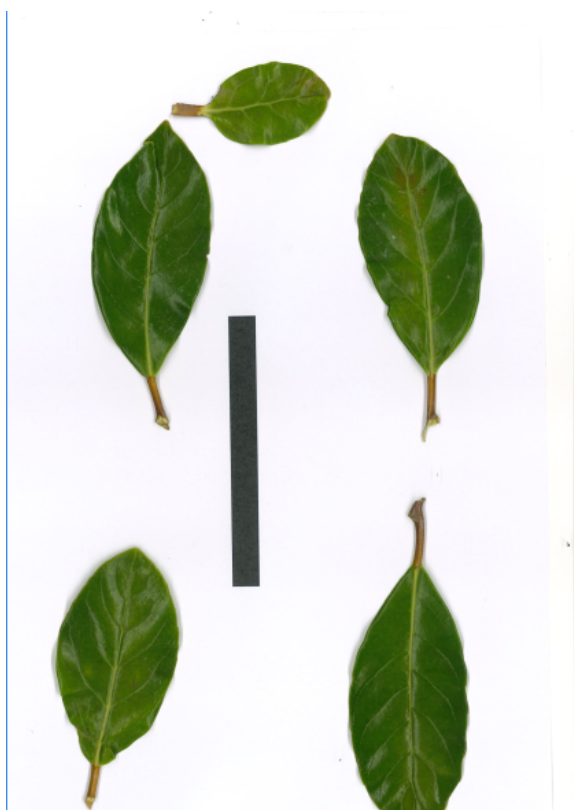


Fig.1 – Leaves image



Fig.2 – Leaf on graph paper



Fig.3 – Paper shapes



Fig.4 – Reference Line

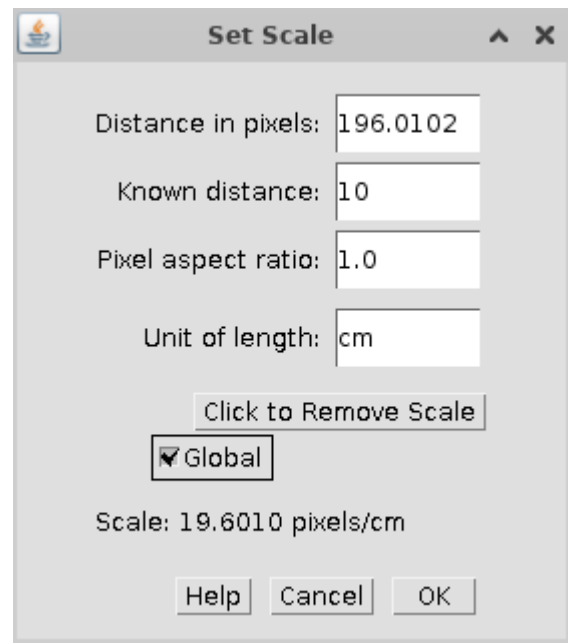


Fig.5 – Set Scale

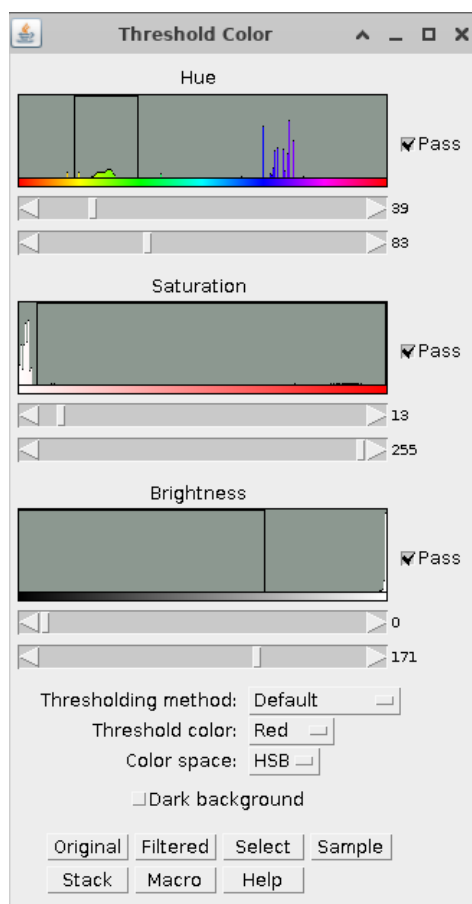


Fig.6 – Color Threshold

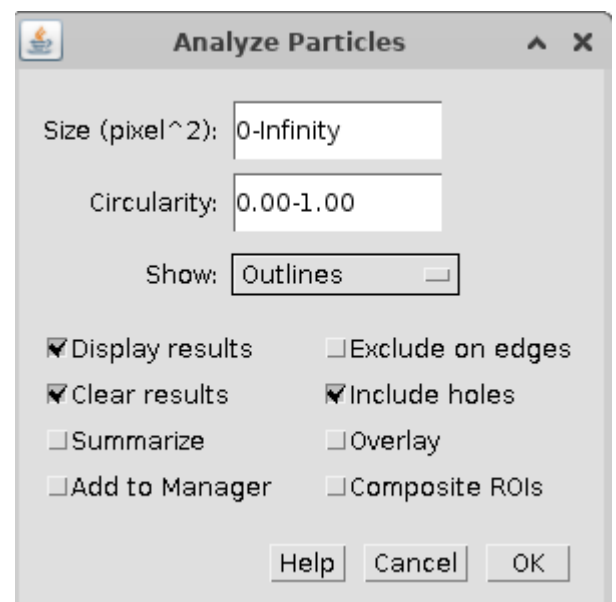


Fig.7 – Analyze particles